WP3 CAPACITY BUILDING WORKSHOP

University of Nairobi

25th-26th October 2022

EU H2020 PROJECT

GA 862848

This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 862848

EWA-BELT

Detection of mycotoxigenic species in plant samples with the portable qPCR Q3 device

Marco Cereda

STMicroelectronics, Italy

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STMicroelectronics (ST): company presentation

October 2022

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We are creators and makers of technology

One of the world’s largest semiconductor companies

48,000 employees of which
8,400 in R&D

Over **80** sales & marketing offices serving over **200,000** customers across the globe

14 manufacturing sites

Signatory of the United Nations Global Compact (UNGC)
Member of the Responsible Business Alliance (RBA)

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As of December 31, 2021

Our strategy stems from key long-term enablers

Smart Mobility

Helping car manufacturers make driving safer, greener, and more connected for everyone

Power & Energy

Enabling industries to increase energy efficiency everywhere and the use of renewable energy

Internet of Things & Connectivity

Supporting the proliferation of smart, connected IoT devices with products, solutions & ecosystems

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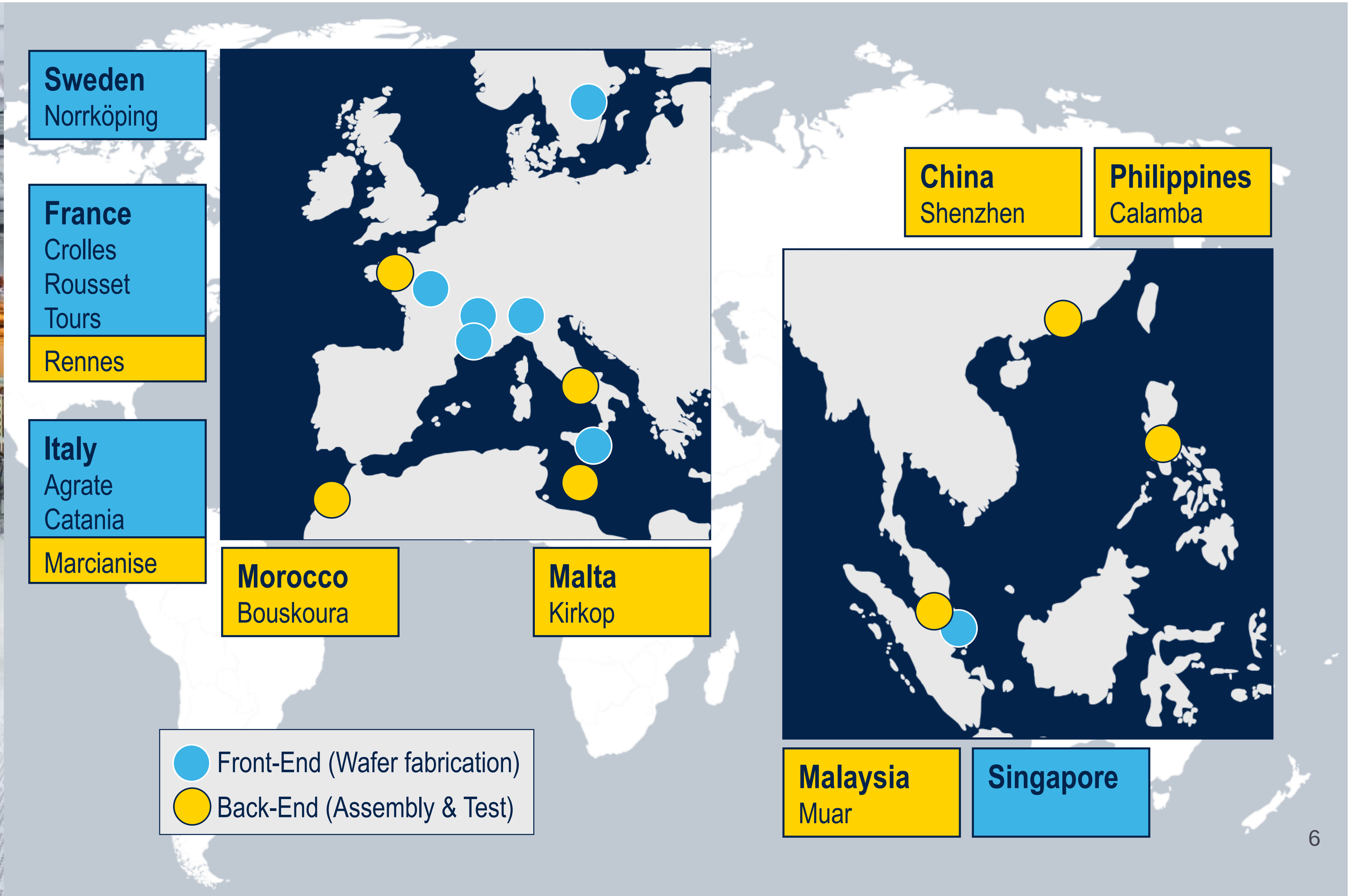
We offer quality, flexibility, and supply security



We are drivers of your innovation

Advanced R&D centers around the world for close collaboration with operations, customers, and partners

~**8,400** people working in R&D and product design

13.5% of revenues invested in R&D in 2021

187 active R&D partnerships

~**18,500** patents

~550 new filings in 2021

Open innovation with startups in **15** Proof of Concept centers

50 startups engaged in our programs

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EWA-BELT: Task 3.2

- T3.2: “Developing and adopting simple, rapid and inexpensive new ICT-based tools for detection and quantification of plant pests and pathogens”, led by ST
- ST has developed a **molecular analysis** workflow to analyze crops of interest, exploiting its “**Q3**” portable device

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Molecular analyses

- Target: **DNA/RNA**
- Plant pests and pathogens can be detected by molecular analyses
- Most used technique: real-time Polymerase Chain Reaction (**qPCR**)
- **Pros**: **sensitivity; specificity; accuracy**
- **Cons**: quite **complex**; **expensive and bulky instrument → tackled in T3.2**
- Workflow:

Sample collection → DNA/RNA extraction → Molecular analysis (qPCR, on Q3)

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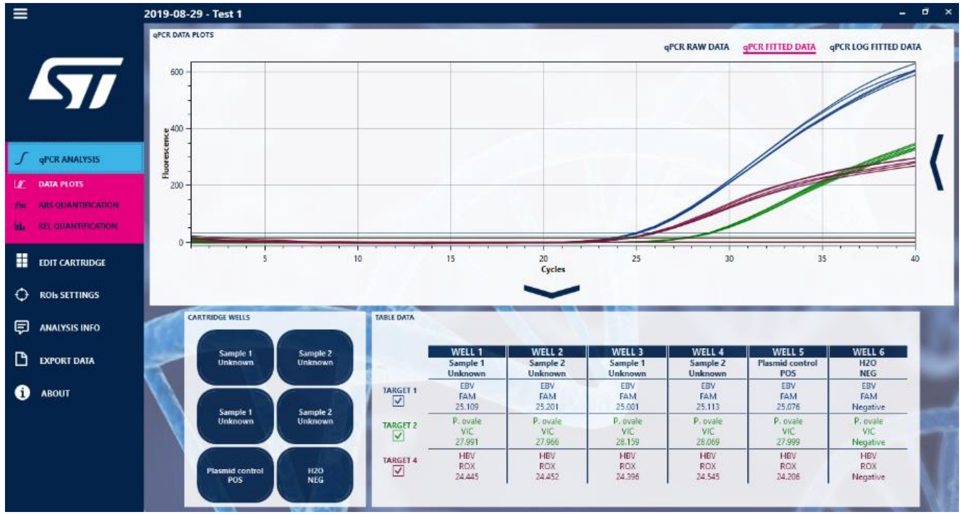
Q3 (BIO2BIT®) platform: overview

- **Compact, Point-of-Care qPCR** platform by ST:
    1. Single-use Lab-on-a-Chip cartridge
    2. Portable instrument (14 x 7 x 8.5 cm; 300 g)
    3. Software
- Advantages vs standard qPCR equipment: **portable, less expensive, easier to use, faster**

1.

2.

3.



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Q3 in EWA-BELT

EWA-BELT

- Background: general-purpose qPCR platform
- Objectives in EWA-BELT:

A. **Task 3.2A**: **development and optimization** of one target application: molecular detection of pathogens in plants of interest for West/East Africa
   - Easy DNA extraction procedure
   - qPCR on Q3 (targets, reagents, protocol)
   - EWA-BELT-dedicated Q3 software, to easily run the qPCR test, whose output files can be uploaded to PLANTHEAD portal

B. **Task 3.2B**: **training to end-users** and field validation **in Africa**

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Task 3.2A_Phase 1

- Development and optimization of different protocols for DNA extraction from **maize flour** (samples provided by UNISS)
- Study of DNA sequences for qPCR detection of **fungal species** of interest, that **can produce mycotoxins**:
  - *Aspergillus flavus, A. ochraceus, A. parasiticus, Fusarium culmorum, F. graminearum, F. proliferatum, F. verticillioides, Penicillium verrucosum*
  - **But some strains** exist that are **genetically unable to produce mycotoxins** (e.g. *A. flavus* strains in Aflasafe®)…
- Setup and optimization of qPCR reaction mixtures
  - Triplex (3 targets): fungus 1, fungus 2, maize DNA sequence as control
- Porting of qPCR onto Q3

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Task 3.2A_Phase 2

- Optimization of protocols for DNA extraction from **flour of different crops** (maize, peanuts, wheat, sorghum, coffee samples by UNISS)
- Study of **fungal** DNA sequences of **mycotoxin biosynthesis genes**:
  - **Ochratoxin A** (OTA)
  - **Aflatoxins** (Afla)
  - **Fumonisins** (Fum)
  - **Trichothecenes** (Tri)
  - **Nivalenol** trichothecene (Niv)
  - 3-acetyldeoxynivalenol trichothecene (**3ADON**)
  - 15-acetyldeoxynivalenol trichothecene (**15ADON**)
  - **Zearalenone** (Zea)

If present in sample, there is a contamination by some **fungus** (which exact species is not important) that for sure is **genetically able to produce** that **toxin(s)**!

- Setup, optimization, and porting to Q3 of qPCR reaction mixtures:
  - Triplex: mycotoxin 1, mycotoxin 2, pan-plant DNA sequence as control

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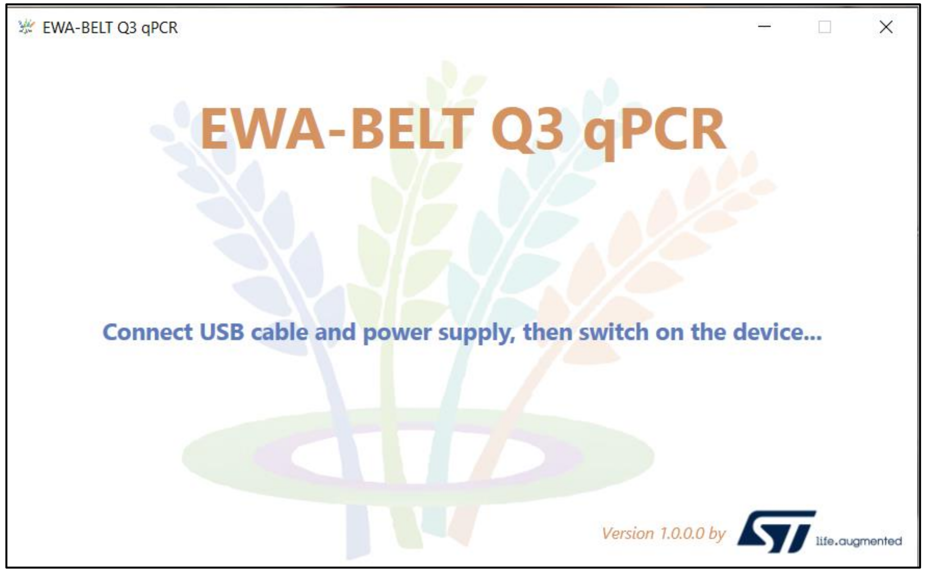
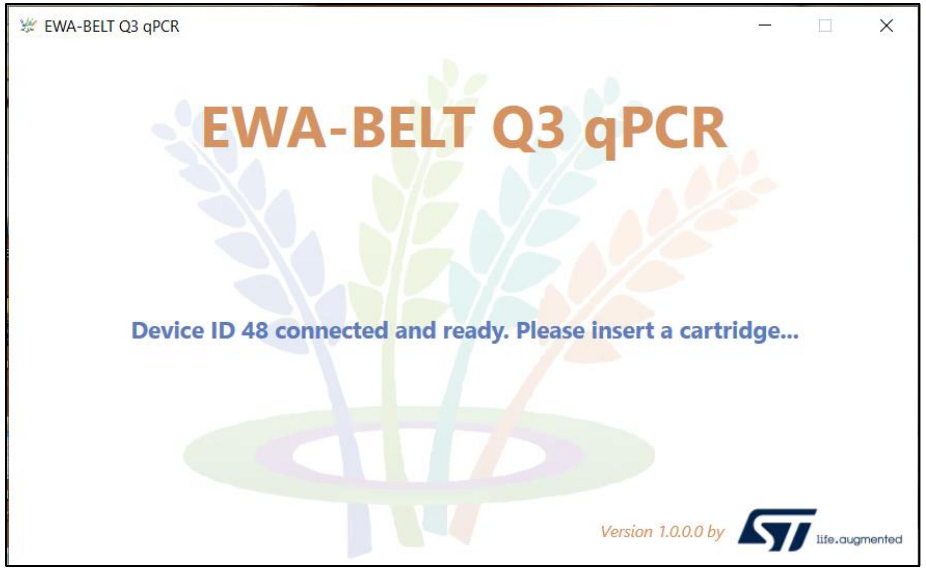
Porting to Q3 on phase 2 (1/3)

- **qPCR** reaction mixtures ported on **Q3** (time-to-result = **45 min**)
- Development and testing of **EWA-BELT-dedicated Q3 software**:
  - **Very easy-to-use**: step-by-step writings to guide the user; all parameters pre-defined
  - Embedded algorithm for raw data interpretation, showing **results easily understandable** by non-expert users
  - **Output** files **uploadable to PLANTHEAD portal**





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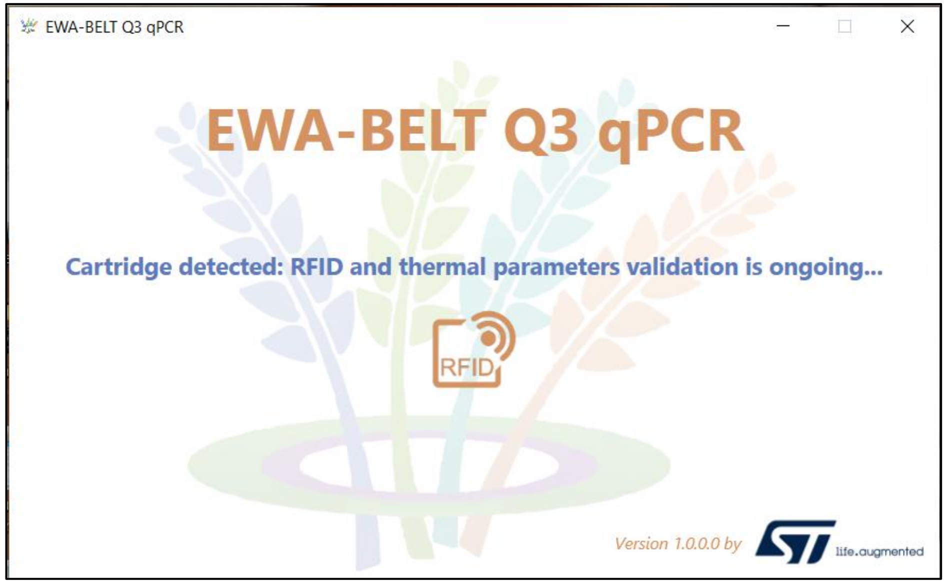
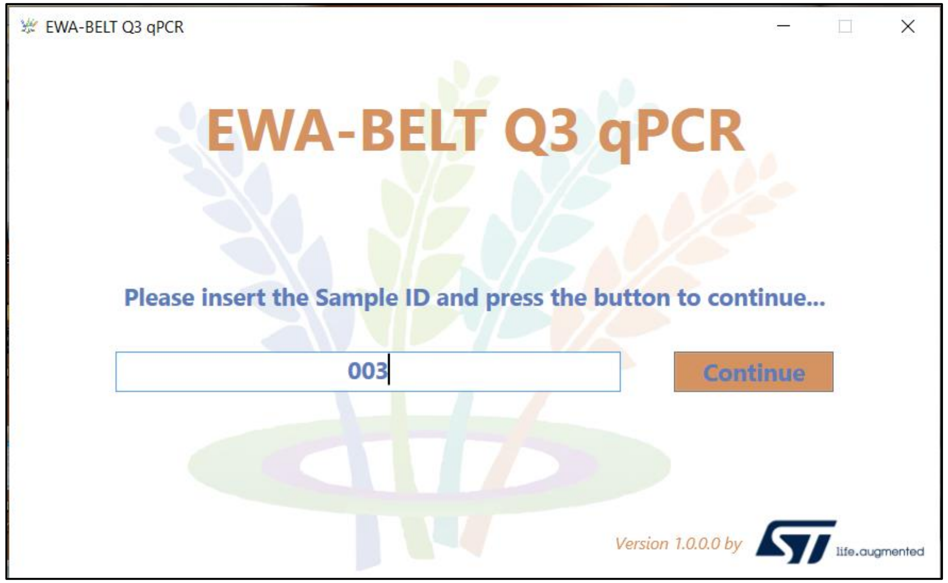
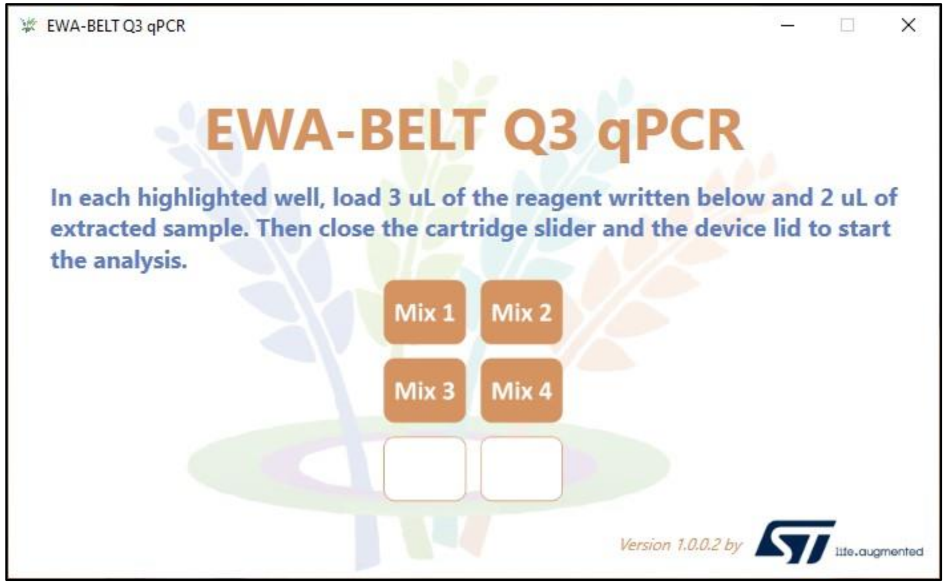
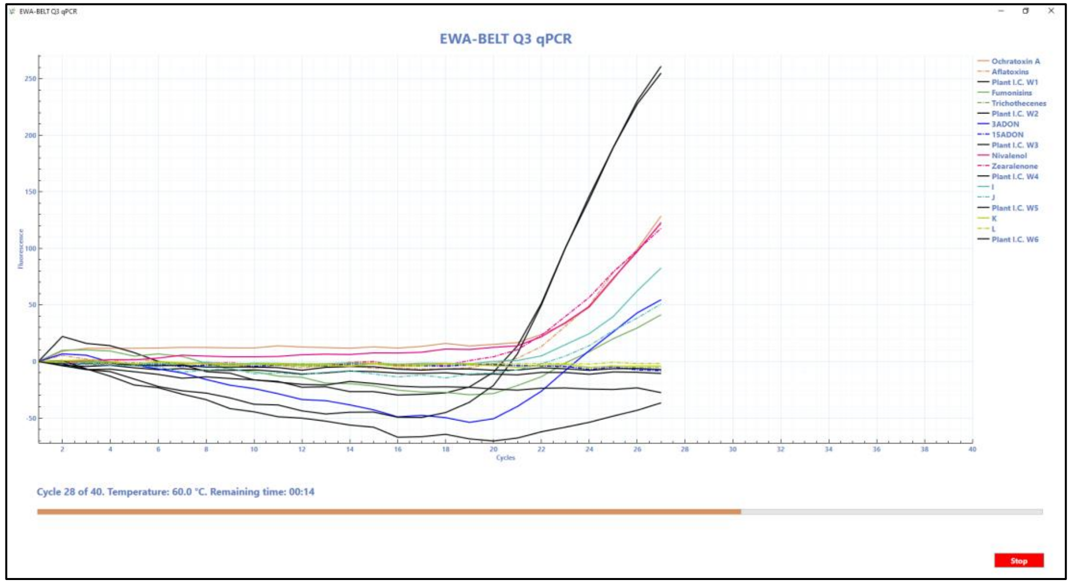
Porting to Q3 on phase 2 (2/3)









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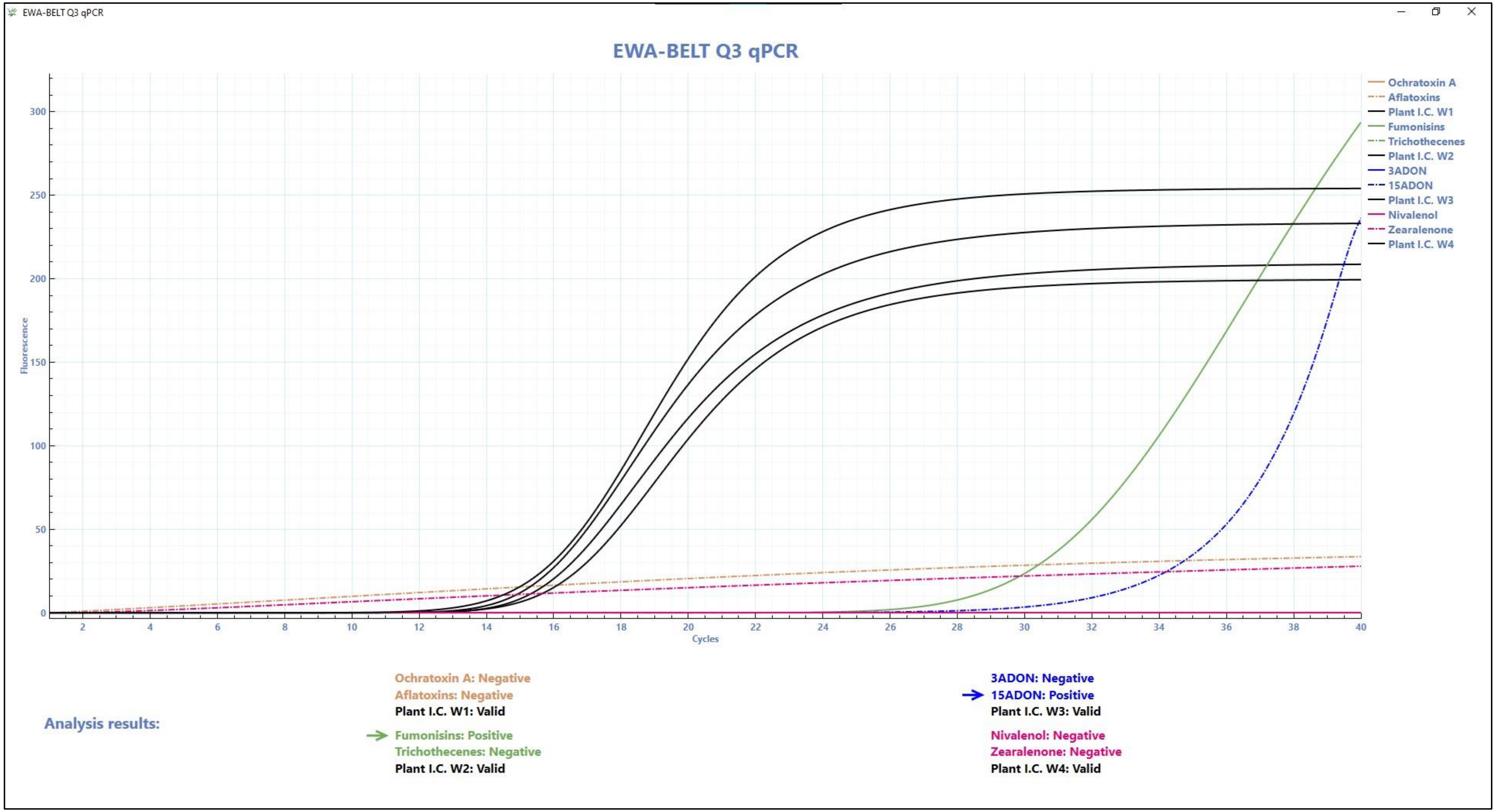
Porting to Q3 on phase 2 (3/3)



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DNA extraction demo

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DNA extraction: basics

- Allows to **purify DNA** from all the other components of a sample
- **Needed before qPCR**, because DNA must be released from cells to be accessible to qPCR reagents; also, qPCR enzyme is inhibited by many natural substances present in samples
- **Plants** are a **tough matrix to purify DNA from**: solid & complex matrix, cells have a resistant cell wall, many qPCR inhibitors (polyphenols, humic acids, polysaccharides)
- Different methods: purely chemical (laborious, toxic reagents), spin column (needs centrifuge), **magnetic beads** (typically needs robot)

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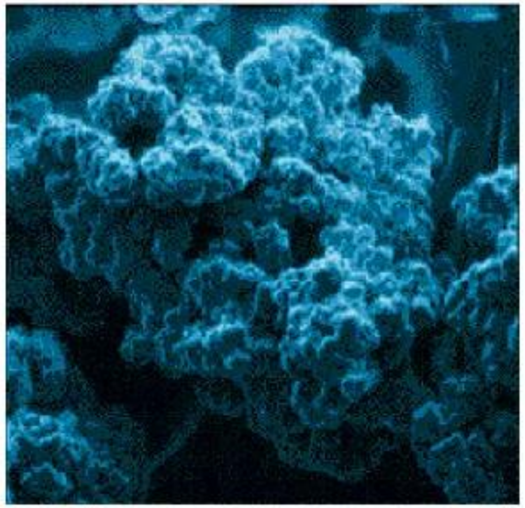
Chosen extraction method: magnetic beads

- Aim: simplest DNA protocol, with very low equipment needed
- Material needed for developed protocol: only two pipettes and one small magnetic rack
- Core technology: paramagnetic, positively-charged microbeads with pores → high surface where DNA can bind
  - Diameter: 0.5 to 8 µm



Scanning electron micrograph of magnetic beads

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DNA extraction: steps

1. **Lysis** → to break cell walls and cell/nucleus membranes, releasing DNA (and all other cellular components) in solution
2. **Binding** (to beads) → appropriate salt/pH conditions make DNA bind to beads
3. **Washes** → to remove all other components, leaving DNA bound to beads
4. **Elution** → to detach DNA from beads, releasing it into solution

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Developed DNA extraction protocol

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0. Sample aliquoting and dissolution:

a. Weigh **200 mg of sample** (flour) into a tube.
b. Add **300 µL water**.

1. Lysis:

a. Add **250 µL *CTAB***. Incubation 5’.
b. Add **250 µL *TLA***. Incubation 5’.
c. Transfer 400 µL supernatant (lysed sample) to a new tube for further processing. Discard the original tube.

2. DNA binding to beads:

a. Add **30 µL *NucleoMag C-Beads*** + **400 µL *MC2*** (Binding Buffer). Magnet 30”. Discard liquid. Remove tube from magnet.

3. Washes:

a. Add **400 µL *MC3*** (Wash Buffer 1). Magnet 30”. Discard liquid. Remove tube from magnet.
b. Add **400 µL *MC4*** (Wash Buffer 2). Magnet 30”. Discard liquid. Remove tube from magnet.
c. Add **400 µL *Ethanol 80%***. Magnet 30”. Discard liquid. Keep tube close to magnet.
d. Gently add **400 µL *MC5*** (Wash Buffer 3). Do not pipette up/down. Incubation 1’ (not more). Discard liquid. Remove tube from magnet.

4. Elution:

a. Add **100 µL *MC6*** (Elution Buffer). Incubation 5’. Magnet 1’. Transfer liquid, containing purified DNA, to a new tube.
b. Store purified DNA at +4°C for up to 24 hours, or at -20°C for longer storage.

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qPCR demo

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Q3 cartridge layout

OTA-Afla	Fum-Tri
3ADON-15ADON	Niv-Zea

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PCR: overview

- Polymerase chain reaction (PCR) is a lab technique for rapidly producing (amplifying) millions-billions of copies of a specific DNA segment, which can then be detected/studied in greater detail
- Short synthetic DNA fragments called primers are designed to select a **target DNA segment** to be **amplified**
- With multiple rounds of DNA synthesis, an enzyme (DNA polymerase) exponentially amplifies that segment
  - $C_n = C_0 * 2^n$, where n is the number of rounds
- Every round consists in **thermal cycle** between ≈ 95°C (denaturation) and ≈ 60°C (primers annealing to target + extension of primers (amplification) by enzyme)
- RNA can also be targeted, adding a second enzyme

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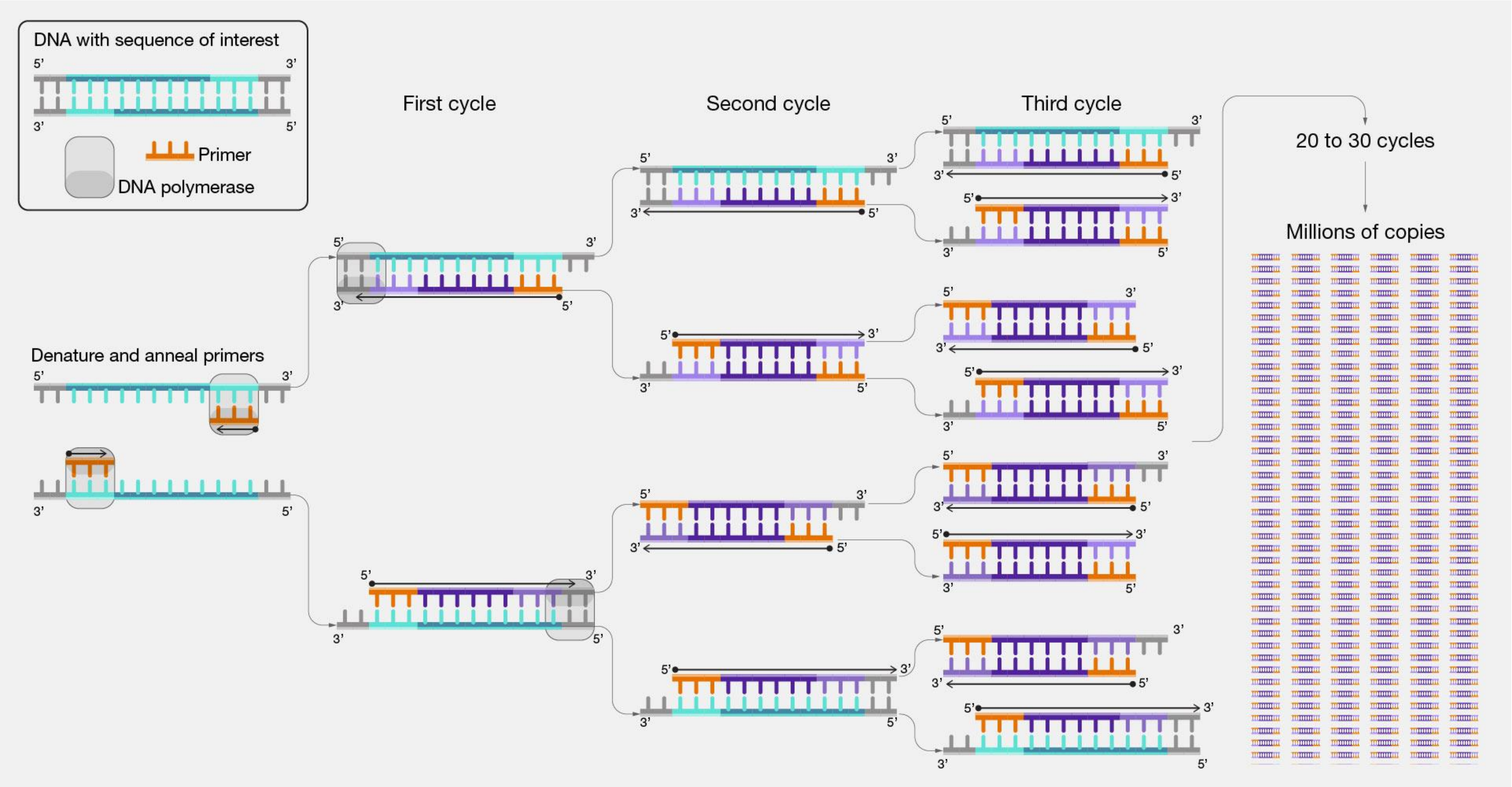
PCR: mechanism



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qPCR: overview

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- In traditional PCR, reaction is analyzed at the end point, when a plateau has typically been reached → No information on initial quantity of target DNA/RNA sequence in the sample, only qualitative answer (yes/no)
- Real-time PCR, or quantitative PCR (**qPCR**) combines PCR with a way to measure amplification while it is occurring
- Measurement during exponential phase provides **quantitative information**. Typically, optical (fluorescence) detection
- Technological requirements for a qPCR platform:

1. Precise and accurate **thermal control** (alternate heating/cooling)
2. **Fluorescence measurement**, typically at different wavelengths

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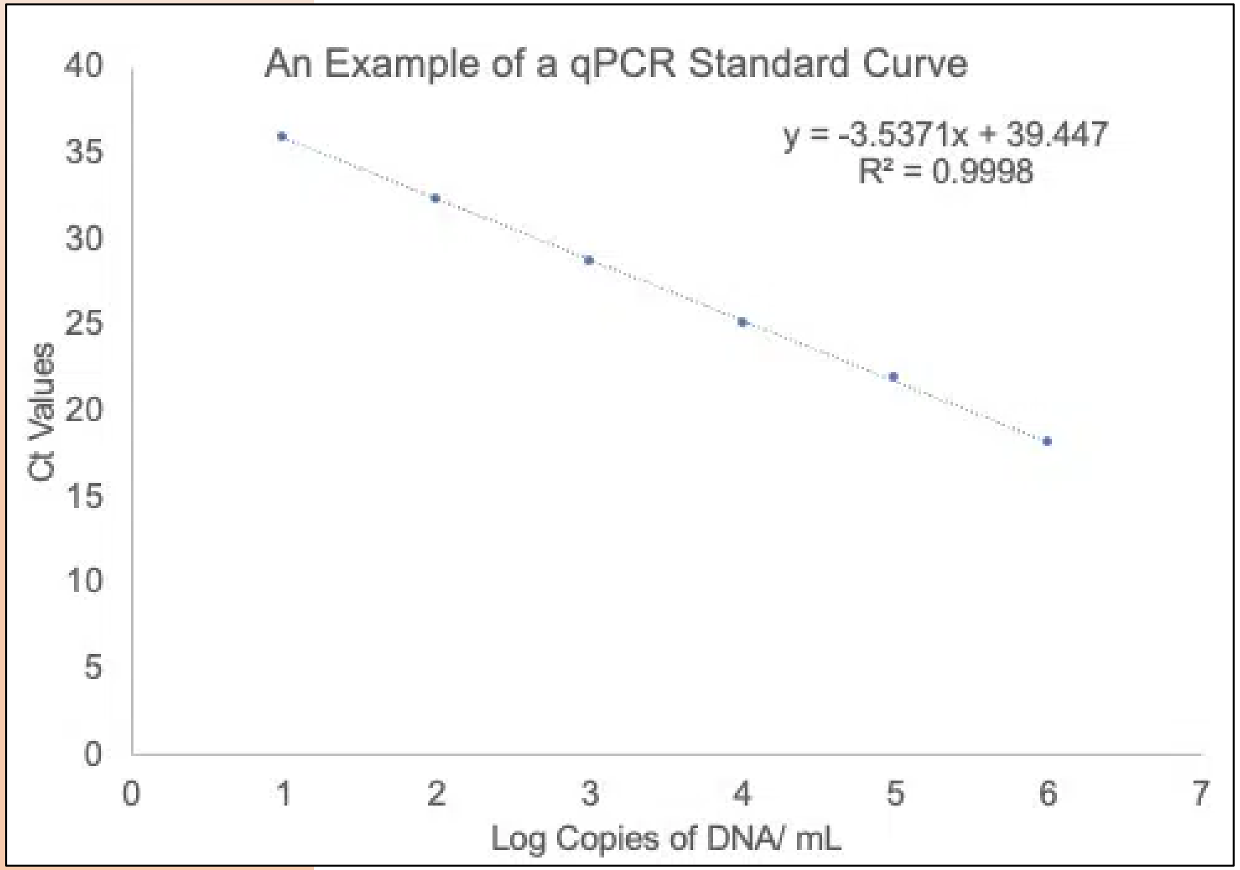
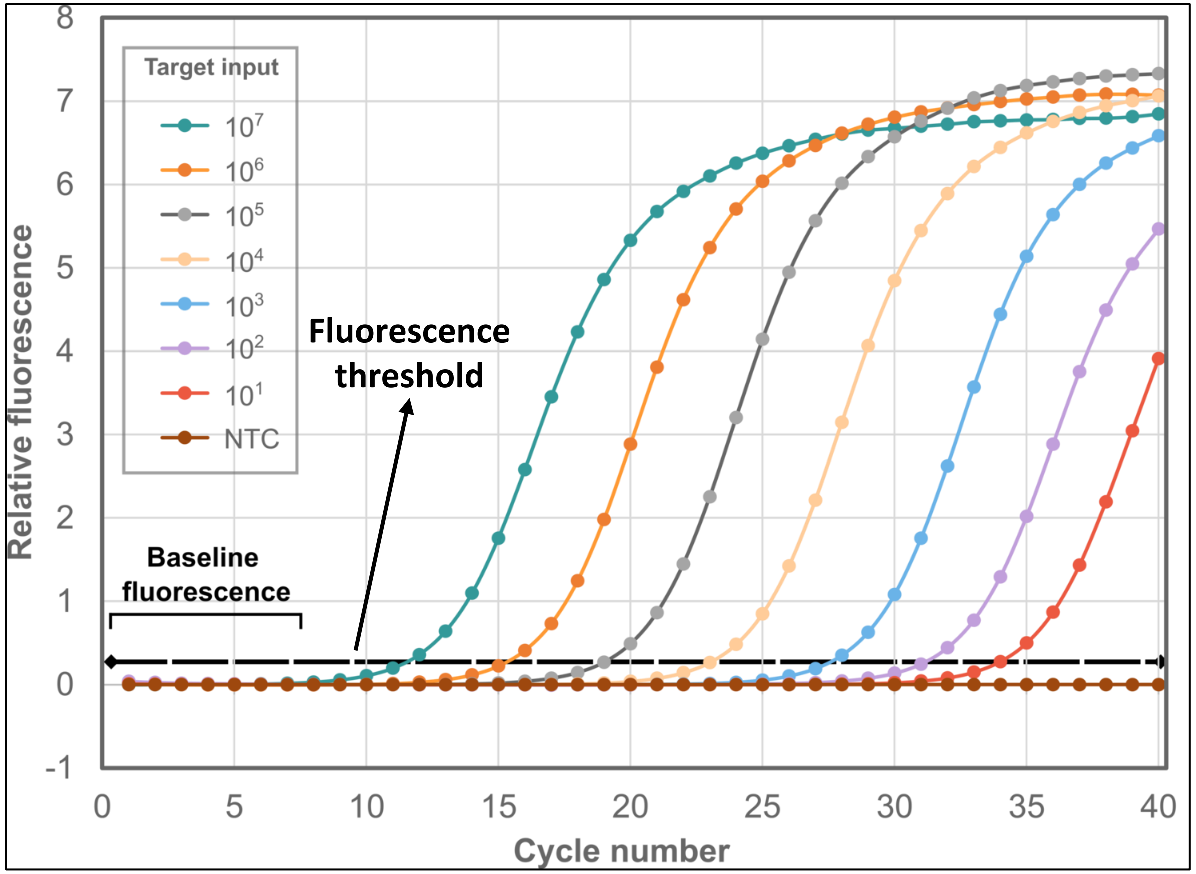
Quantification in qPCR

- A fluorescence threshold is defined in the exponential phase; the moment when fluorescence overcomes this threshold is called **Cq** (or Ct) value for a certain sample → main output of qPCR
- Cq is inversely proportional to the Log of target DNA/RNA initial quantity





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  1. Single-use Lab-on-a-Chip cartridge
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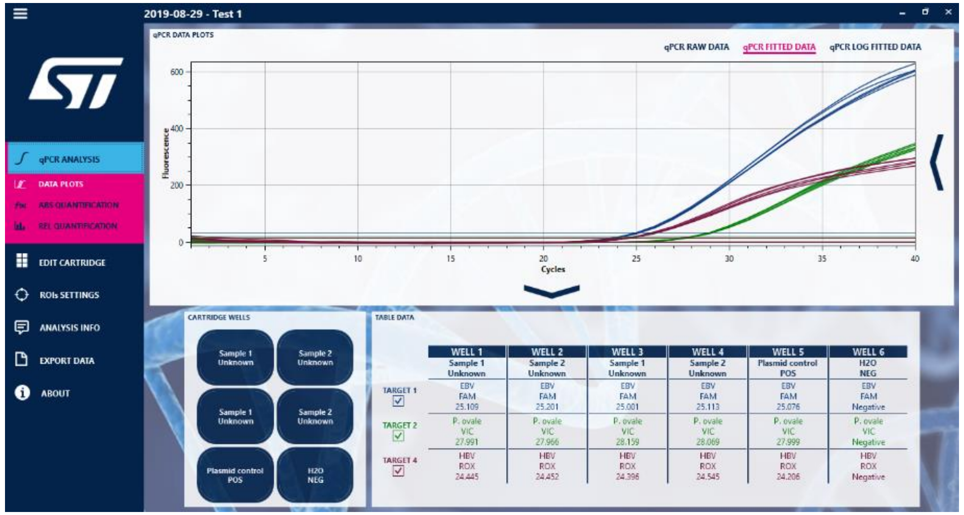
  3. Software
- Advantages vs standard qPCR equipment: **portable, less expensive, easier to use, faster**

3.



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Q3 platform elements: cartridge

- Six reaction chambers, 5 µL each
- Key element is a **silicon chip**, manufactured by ST Italy, integrating thermo-resistive heater and temperature sensor for precise and accurate **thermal control**
- Silicon chip assembled with plastics, wax and RFID tag to obtain cartridge

- Special **wax** avoids evaporation of reagents during thermal cycling, and seals reagents at the end
- **RFID tag**, writable/readable by Q3 instrument and all NFC-enabled devices, allows robust traceability and anti-fraud protection

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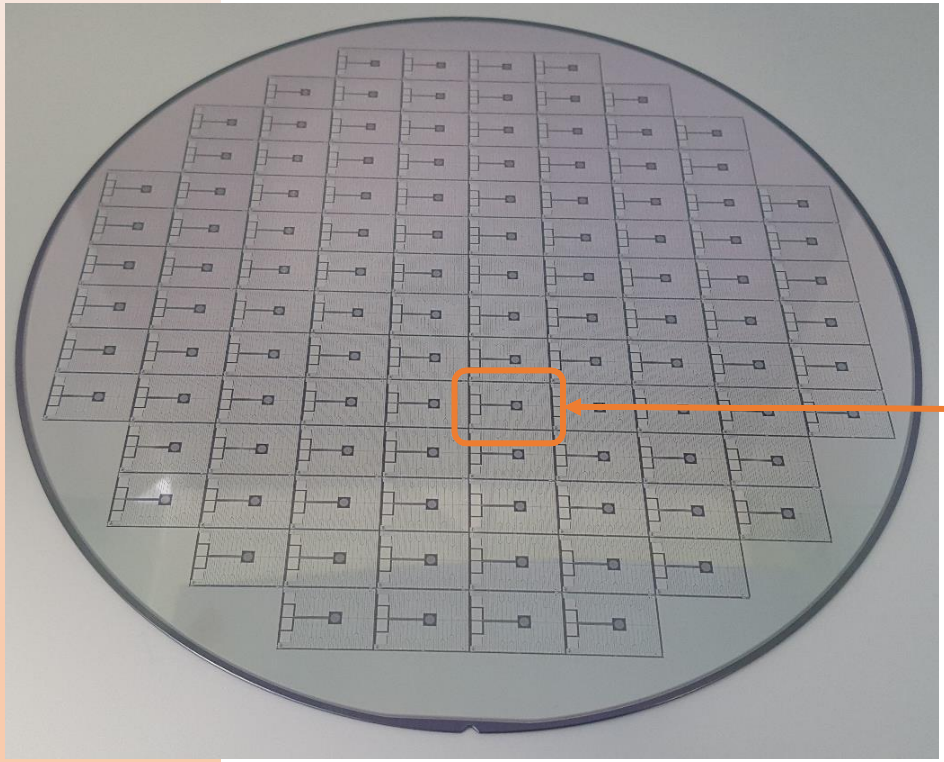
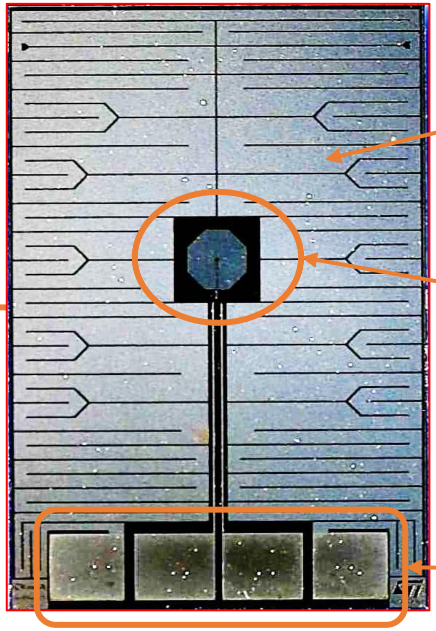
Cartridge silicon chip: metal side

- Chips are produced in parallel onto an 8” silicon wafer





Distributed metal resistor acting as **heater**

Concentrated metal resistor acting as **temperature sensor**

Contacts to Q3 instrument

- Silicon is a good thermal conductor

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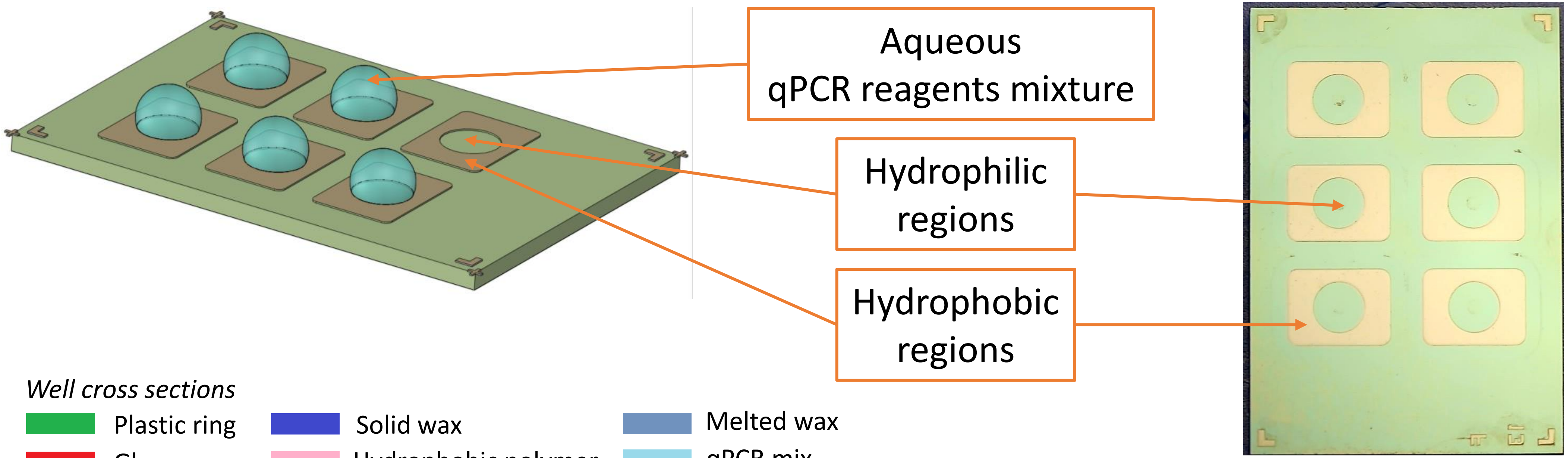
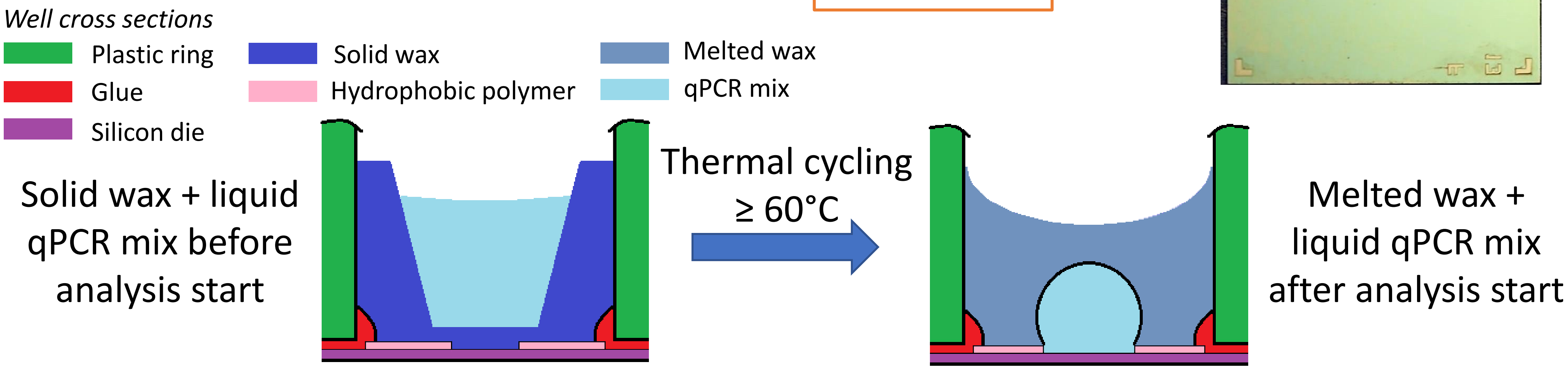
Cartridge silicon chip: reagents side

- Hydrophilic regions: silicon oxide
- Hydrophobic regions: polymer patterned through photolithography





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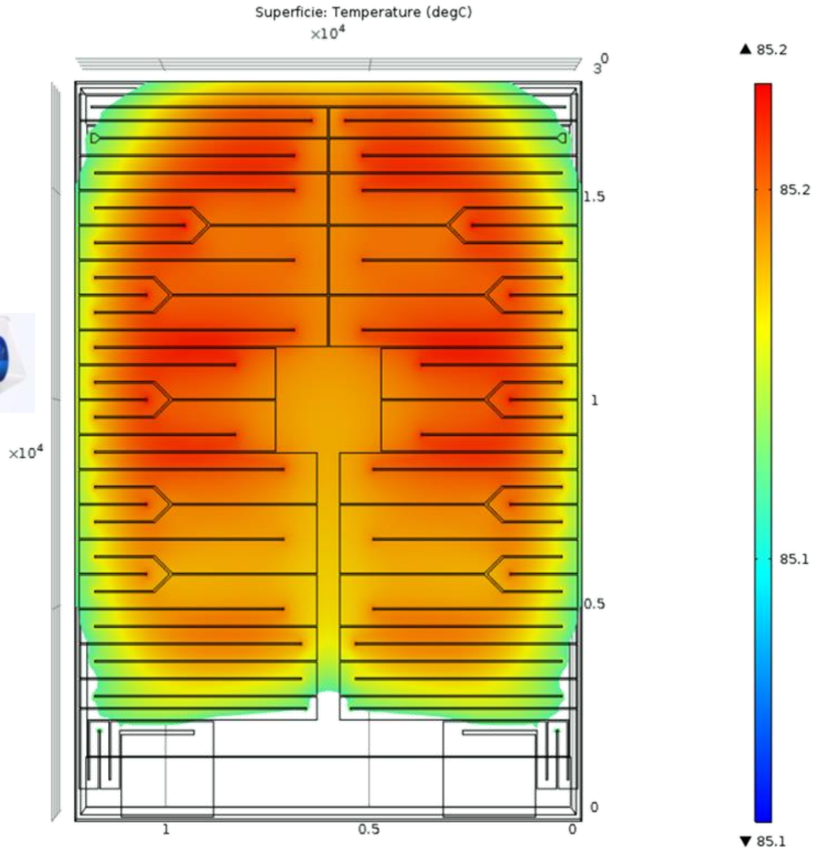
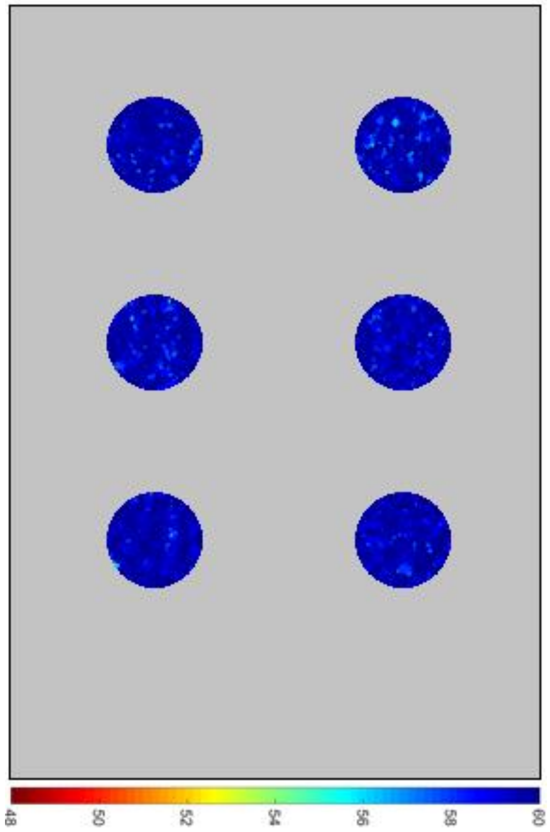
Chip temperature distribution

Simulation of temperature distribution @ 85°C



$\Delta T = 0.1°C$

Characterization with thermo-sensitive liquid crystal @ 60°C



$\Delta T = 0.13°C$

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Q3 platform elements: instrument

1. Drives the cartridge allowing the thermal cycling required by qPCR
2. Acquires **fluorescence signals** coming from cartridge during reaction, on 3 different optical bands
3. Allows communication with host through USB or Bluetooth

- Only 14 x 7 x 8.5 cm and weighs barely 300 g
- Internal memory for backup storage of up to 350 tests
- RFID tag reading/writing and cryptographic scheme for data protection and chip cloning prevention
- Motorised lid and touch button with multi-color status LED

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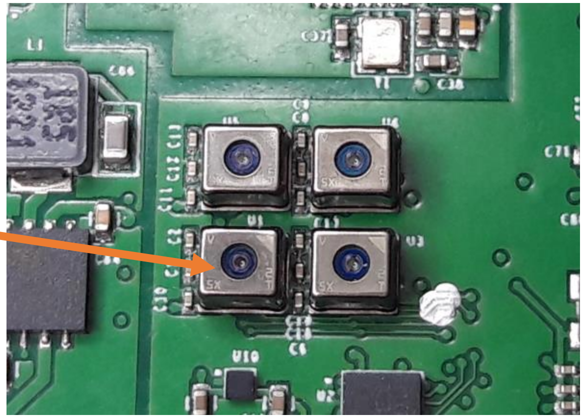
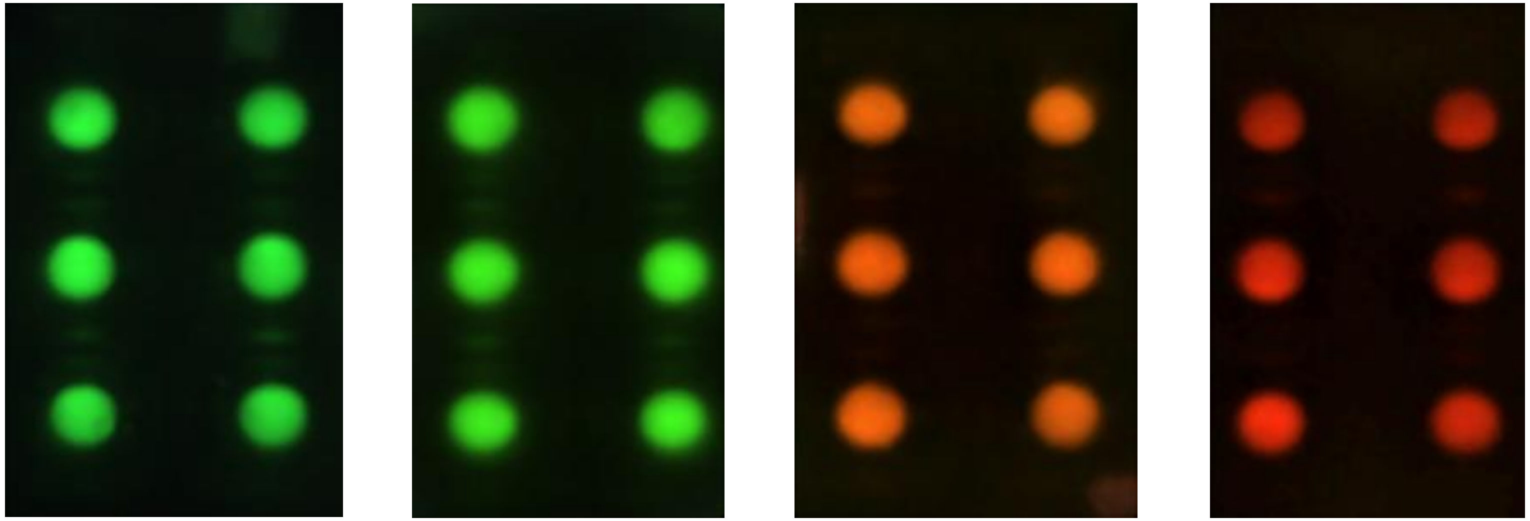
Instrument: optics

Compact fluorescence microscope, with no moving parts:

- 4 LEDs illuminating the cartridge
- 4 ST CMOS image sensors, closely spaced, looking at the same area
- Each sensor and LED is behind a different optical filter



Images obtained from each optical channel:



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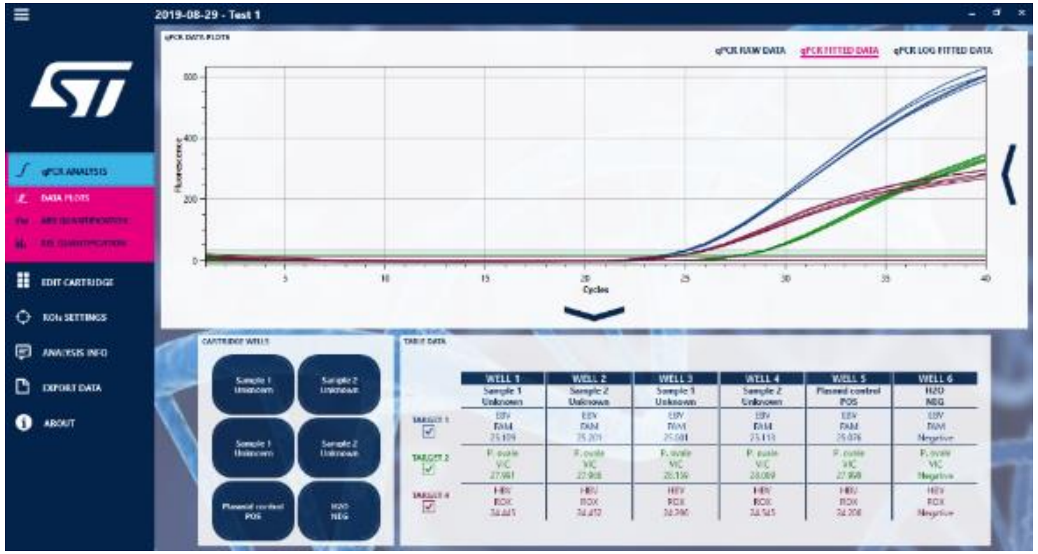
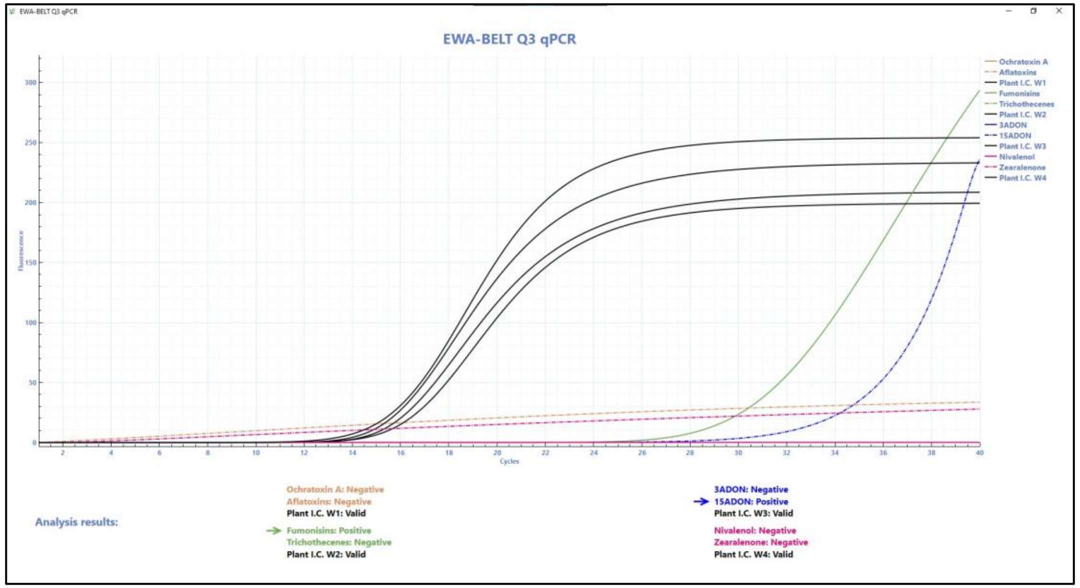
Q3 platform elements: software

1. Background: **general-purpose software**
   - Full protocol flexibility
   - General data analysis (Cq calculation)



2. Developed for **EWA-BELT**: “closed”, dedicated **software**
   - Pre-defined protocol
   - Writings guide the user step-by-step
   - Dedicated data analysis (positive/negative for each mycotoxin)



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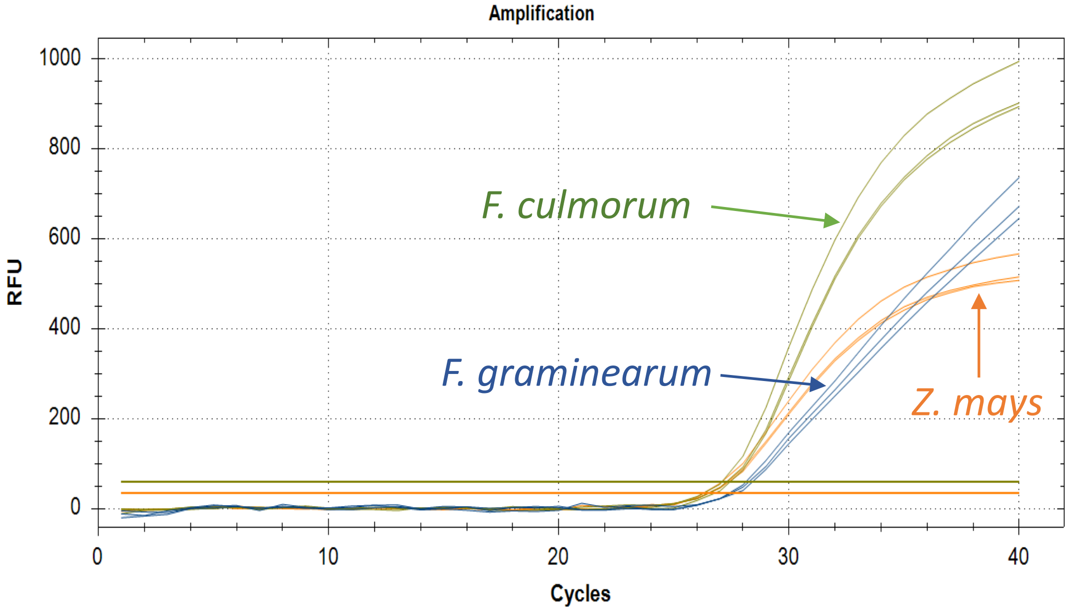
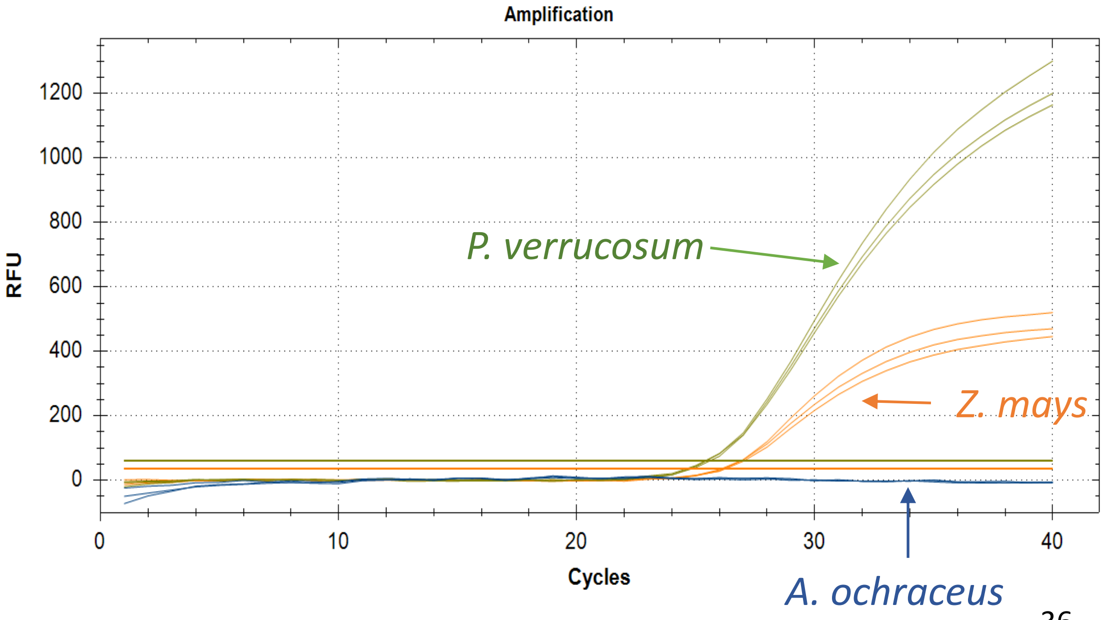
Results of phase 1 (1/2)

- **Simple DNA extraction protocol** developed and tested on maize: only two pipettes and one magnet needed; **≈ 40 min**
- **Two qPCR reaction mixtures optimized**, targeting *F. culmorum, F. graminearum, A. ochraceus, P. verrucosum*
- Results on gold standard qPCR equipment:

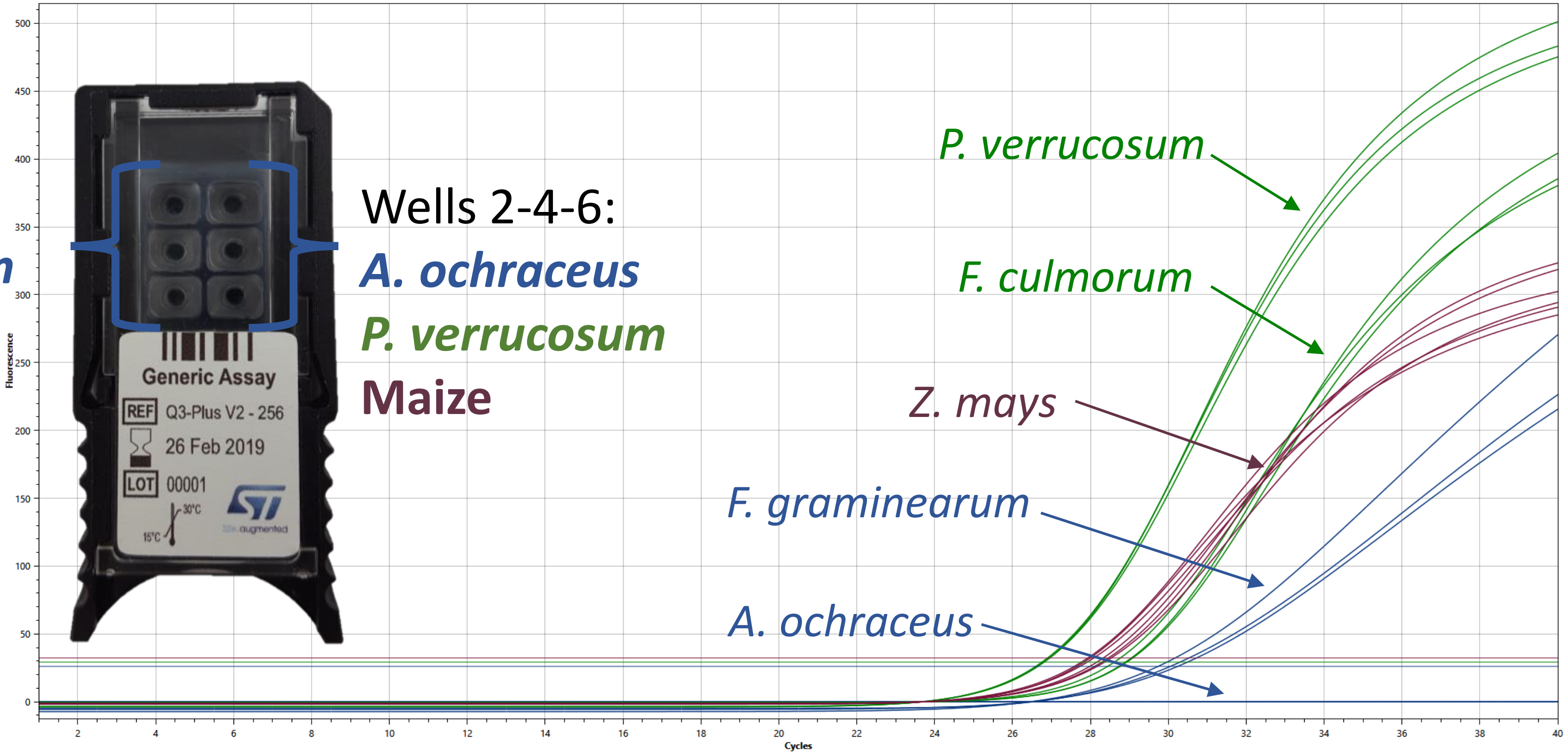




Results of phase 1 (2/2)

- **qPCR** reaction mixtures ported on **Q3** (time-to-result = **45 min**)

Wells 1-3-5:
F. graminearum
F. culmorum
Maize



- **Fungi clearly detected**
- **Good consistency** of **replicates**

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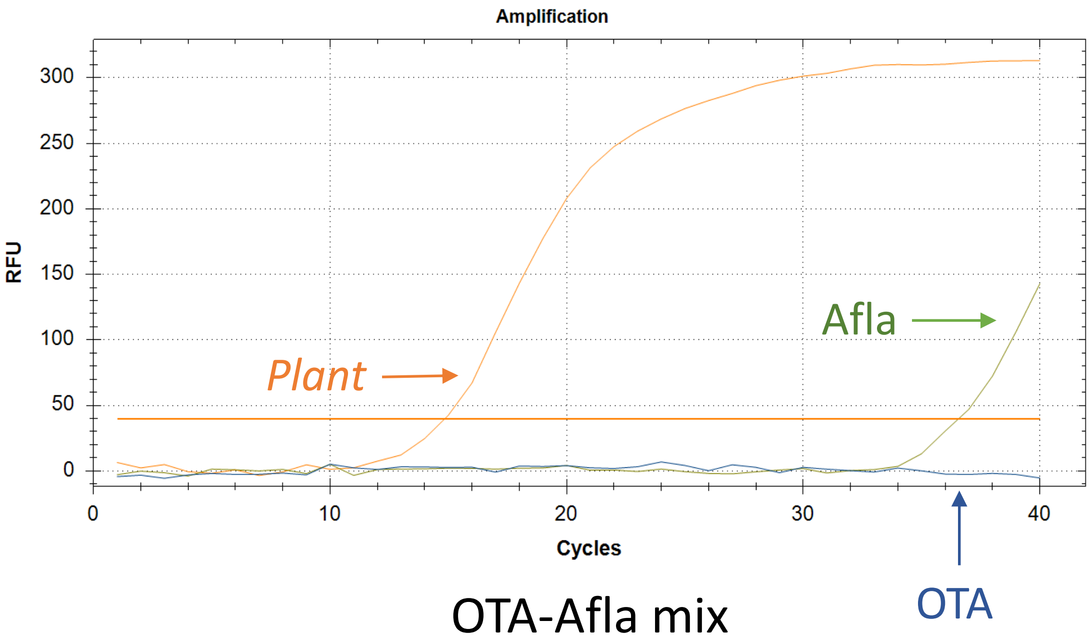
Preliminary results of phase 2 (1/3)

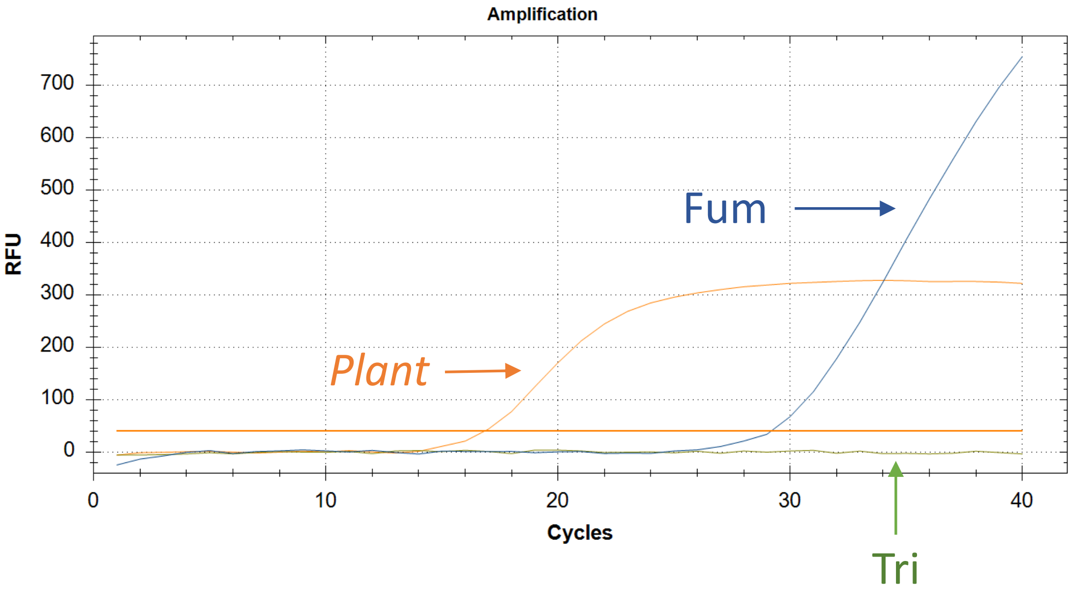
- Simple DNA extraction protocols developed and tested on maize and peanut: only two pipettes and one magnet needed; ≈ **40 min**
- **Four qPCR mixes developed**, targeting genes for 8 mycotoxins
- Results on gold standard qPCR equipment:

A. flavus-contaminated Kenyan peanut



OTA-Afla mix

F. proliferatum-contaminated maize



Fum-Tri mix

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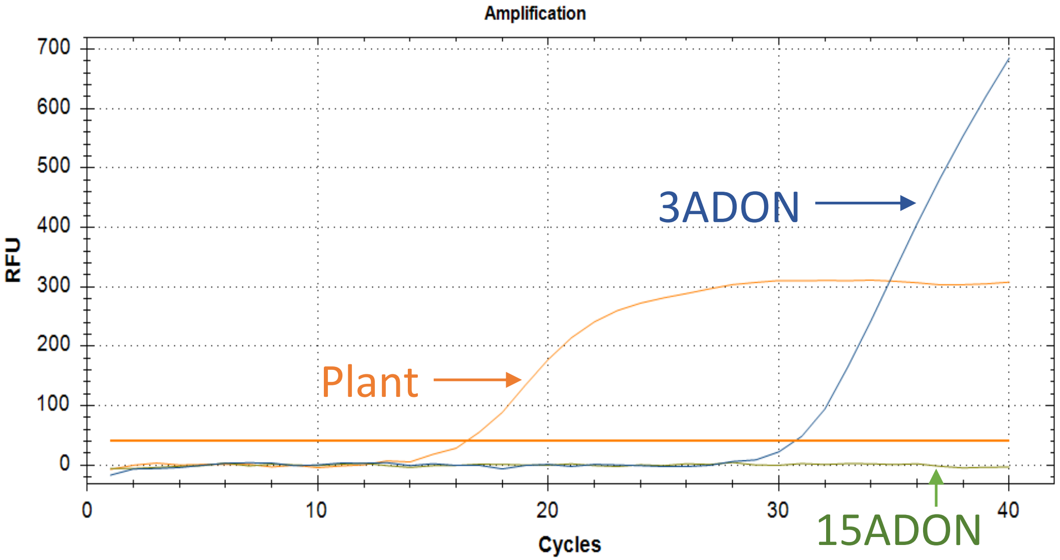
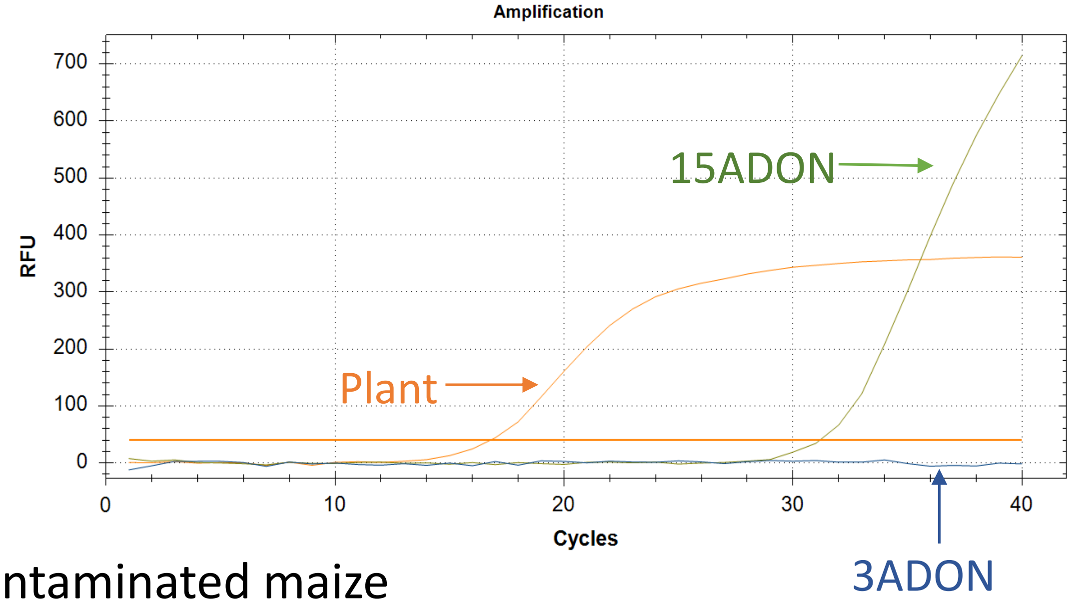
Preliminary results of phase 2 (2/3)

ADON mix

F. culmorum-contaminated maize

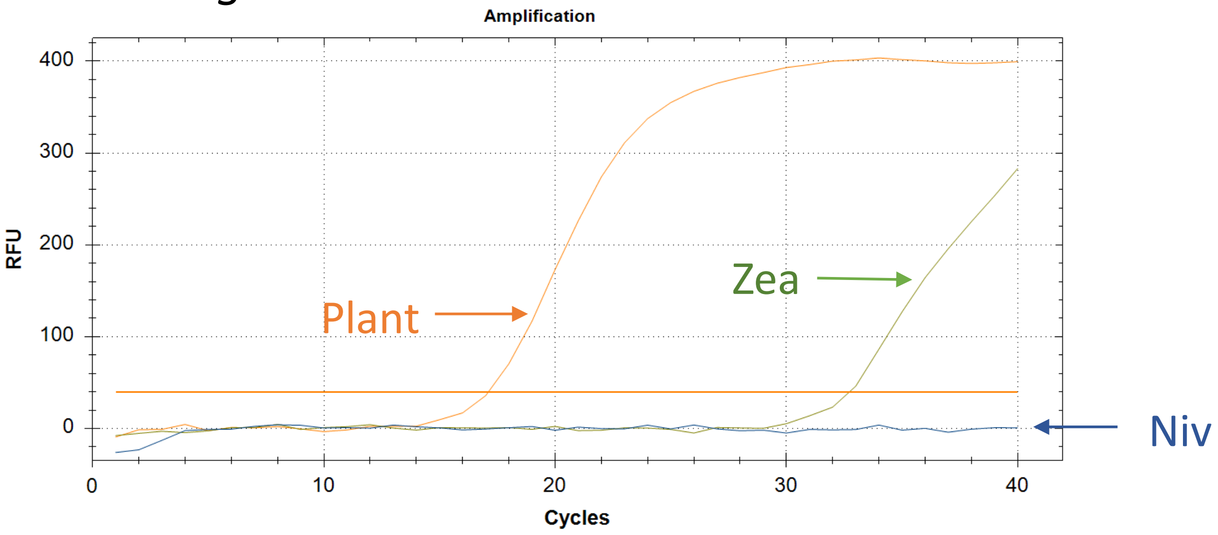


F. graminearum-contaminated maize



Niv-Zea mix

F. graminearum-contaminated maize



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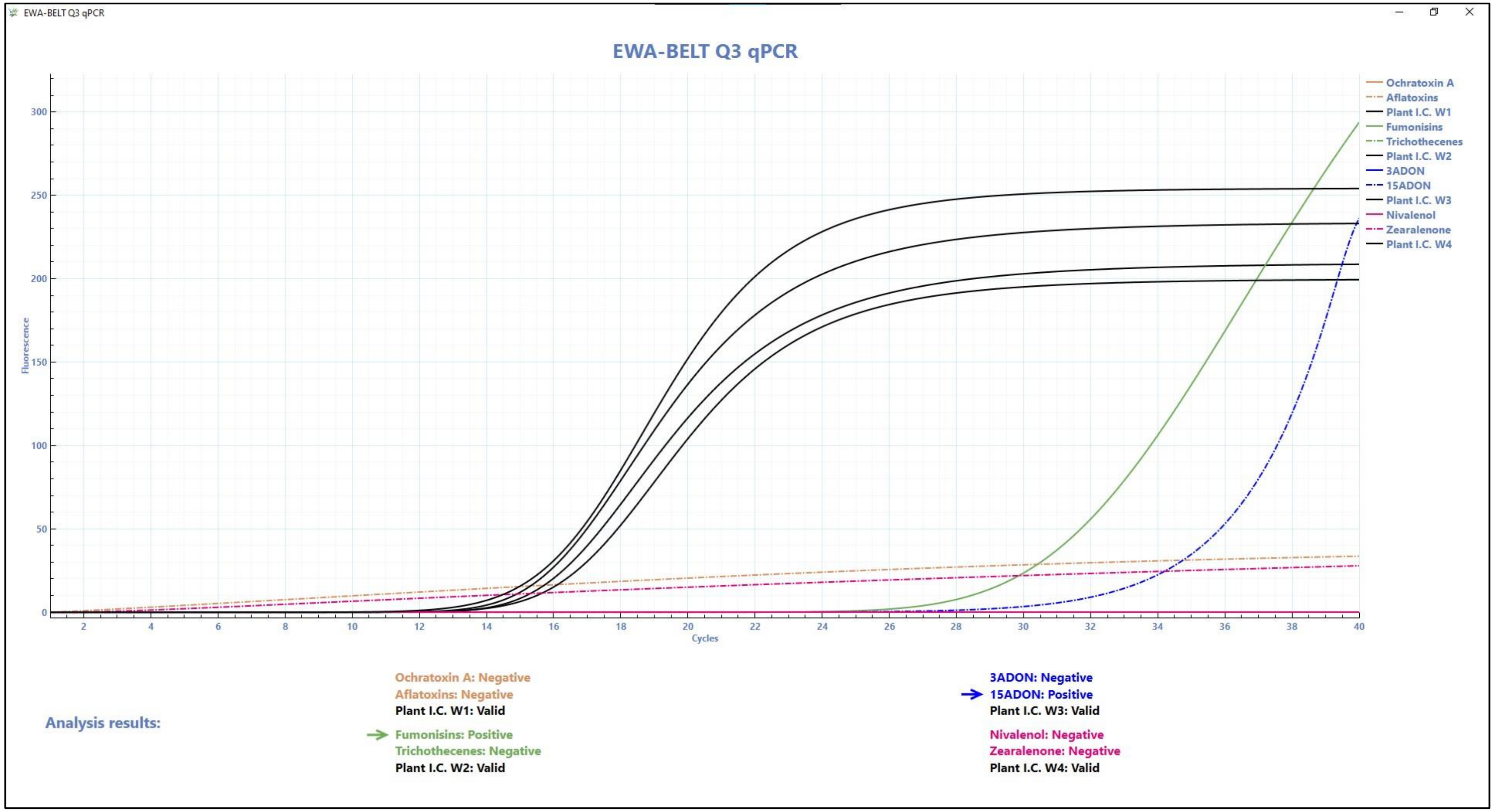
Preliminary results of phase 2 (3/3)

- **qPCR** reaction mixtures ported on **Q3** (time-to-result = **45 min**)
- Tests run with **EWA-BELT-dedicated Q3 software**

F. proliferatum-contaminated maize



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EWA-BELT

Next steps

- Optimization of DNA extraction protocols for different plants (wheat, sorghum, coffee)
- Further optimization of qPCR detection of some mycotoxins
- Fine-tuning of algorithm embedded in EWA-BELT software for raw data interpretation

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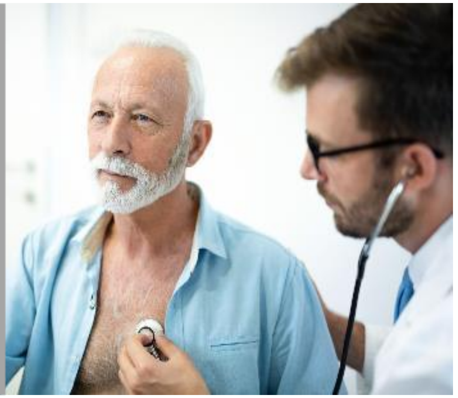
Possible applications of Q3

Infectious Diseases

COVID-19 [1], Sepsis [1], Malaria [2], Ebola [3]

Cardiology

Antiplatelet drug metabolism [4,5]



Pediatrics

Meningitis, Pertussis [6]

Gynecology

HPV, Thrombophilia

Oncology

Pharmaco-genetics

Agro-food

Parasites in mollusks [7], meat identification, **EWA-BELT**

(1) https://alifax.com/assets/Uploads/STM-BRO001-MOLECULAR-MOUSE-03-06-17062022-EN.pdf

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Thank you!

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